

A Project Stage – II

on

“SMART MANHOLE”

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
in
ELECTRONICS AND COMMUNICATION ENGINEERING
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Department of Electronics and Communication Engineering
BVRIT HYDERABAD College of Engineering for Women

(Approved by AICTE, New Delhi and Affiliated to JNTUH, Hyderabad)

Accredited by NBA and NAAC with A Grade

Bachupally, Hyderabad – 500090

2025-26

DECLARATION

We hereby declare that the work described in this report, entitled “**SMART MANHOLE**” which is being submitted by us in partial fulfilment for the award of the degree of **Bachelor of Technology** in the Department of **Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women**, affiliated to **Jawaharlal Nehru Technological University Hyderabad**, Kukatpally, Hyderabad – 500085 is the result of original work carried out by us under the guidance of **Dr. J.Naga Vishnu Vardhan, Professor, ECE and Dean Academics**

This work has not been submitted for any Degree/Diploma of this or any other institute/university to the best of our knowledge and belief.

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Certificate

This is to certify that the major/mini project report, entitled “**SMART MANHOLE**” is a record of bonafide work carried out by **Kalluri Hansika 22WH1A0430, Dhatri Sri Singu 22WH1A0410, K Harika 22WH1A0434, S Sanika 22WH1A0451** in partial fulfilment for the award of the degree of **Bachelor of Technology** in the department of **Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women**, affiliated to **Jawaharlal Nehru Technological University Hyderabad, Kukatpally, Hyderabad – 500085**.

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ABSTRACT

Manholes play a vital role in managing underground utilities, such as sewage, storm drains, and water supply systems, but they also present significant risks during heavy rainfall or blockages, including public safety hazards and environmental contamination. This paper proposes a smart manhole monitoring system that leverages Internet of Things (IoT) technology to address these challenges. The system utilizes float sensors to monitor water levels and detect potential overflows, triggering immediate alerts to municipal authorities for timely intervention. Additionally, LED indicators placed around the manhole serve as visual warning signals for missing or damaged caps, preventing accidents. Unlike traditional systems that depend on GSM modules, the proposed solution uses LoRa communication, reducing reliance on telecom networks and minimizing maintenance costs. Power is supplied via nearby streetlights, while protective enclosures ensure hardware durability in harsh environmental conditions. This innovative system enhances proactive urban infrastructure management by combining real-time monitoring, automation, and smart communication. By improving public safety, reducing environmental hazards, and supporting sustainable smart city initiatives, the smart manhole system offers a cost-effective and scalable solution for modern urban management.

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1. Introduction

1.1 Overview of Manhole Systems

Manholes are vertical access points that connect the ground surface to underground utility networks. They provide entry for inspection, maintenance, cleaning, and repair of systems such as sewage lines, stormwater drains, electrical cables, and water pipelines. A typical manhole consists of a concrete or metal chamber with a removable cover placed at ground level. These structures ensure that technicians can safely access underground infrastructure without excavation.

1.2 Importance of Underground Utilities

Underground utilities are like the hidden lifelines of a city. We may not see them every day, but we depend on them constantly. From the moment we wake up and use water at home to the time we switch on lights or use the internet, these systems are quietly working beneath the ground to support us.

They make sure wastewater is safely carried away, keeping our surroundings clean and healthy. During heavy rains, they help drain excess water and prevent flooding on roads. They provide a steady supply of clean water to our homes and safely protect electrical and telecommunication cables underground.

Safe housing of telecommunication and electrical cables. Since these utilities are hidden beneath the surface, they remain protected from weather and human interference. Their efficient functioning is essential for public health, environmental stability, and overall urban development. Any fault or overflow in these systems can disturb daily life and require immediate attention.

1.3 Causes of Manhole Overflow

Manhole overflow occurs when the internal water level rises beyond the system's capacity. The major causes include:

- **Heavy Rainfall:** Excess storm water enters drains and increases pressure inside the manhole.

- **Blockages:** Accumulation of plastic, debris, or silt restricts the flow inside pipelines.
- **Sewer Line Damage:** Cracks, pipe collapse, or tree root intrusion reduce flow and cause backpressure.
- **Poor Maintenance:** Irregular cleaning allows waste buildup.
- **Illegal Connections:** Excess load from unauthorized drainage pipes can exceed the system limit. These issues lead to rising water levels that can result in overflow onto roads and public areas.

1.4 Effects on Public Safety

Manhole overflow or missing/broken manhole covers create serious safety risks:

- Pedestrians, cyclists, or vehicles may fall into an uncovered manhole.
- Slippery roads due to sewage water increase the chances of accidents.
- Open manholes at night become nearly invisible, posing greater danger.
- Overflowing wastewater can spread harmful bacteria, leading to health hazards. Therefore, ensuring the safety of manholes is crucial for preventing accidents and protecting the public.

1.5 Environmental Impact

Manhole overflow releases untreated sewage or contaminated stormwater into the environment. This causes:

- **Water contamination** of lakes, rivers, and groundwater.
- **Soil pollution** due to chemicals and waste seeping into the ground.
- **Bad odour** and unhygienic surroundings.

- **Spread of harmful microorganisms** that can cause diseases. Such impacts degrade environmental quality and pose long-term threats to local ecosystems.

1.6 Infrastructure Damage Due to Overflow

Excessive pressure and forceful water discharge from manholes can damage surrounding structures, such as:

- **Road surfaces**, causing cracks, potholes, and erosion.
- **Footpaths** and nearby buildings due to water seepage.
- **Pipelines** inside the underground network.
- Frequent overflow increases maintenance costs for municipal bodies and reduces the lifespan of public infrastructure.

1.7 Need for Smart Monitoring Solutions

Traditional manhole systems rely on manual inspection, which is slow, inefficient, and often detects issues only after damage occurs. With rapid urban growth and increasing rainfall intensity, early detection and preventive action are essential. Smart monitoring solutions integrate sensors, communication systems, and alert mechanisms to:

- Detect rising water levels before overflow
- Notify authorities in real time
- Identify broken or missing manhole covers
- Improve public safety and reduce accidents

2. Literature Survey

2.1 Existing Monitoring Techniques

The development of Smart Manhole Monitoring Systems using microcontrollers and Internet of Things (IoT) technologies has become an important area of research in modern urban infrastructure management. Many researchers have proposed intelligent monitoring systems to improve the safety and efficiency of underground drainage systems.

Traditional manhole monitoring methods rely heavily on manual inspection, which is time-consuming and often ineffective in detecting problems such as open manhole covers or water overflow. To overcome these limitations, IoT-based monitoring systems have been introduced to provide continuous and real-time monitoring of manhole conditions. In recent years, researchers have explored the use of microcontroller platforms such as NodeMCU and Arduino combined with IoT sensors to detect various parameters related to manhole safety. Sensors such as IR sensors, water level sensors, Float sensors, and accelerometer sensors are commonly used to detect manhole cover displacement, water level rise, or drainage blockages. These sensors collect environmental data and transmit it to a central monitoring system, enabling authorities to take immediate action in case of abnormal conditions. Communication technologies play a critical role in the performance of IoT-based monitoring systems. Traditional short-range communication methods such as Wi-Fi and Bluetooth have limitations in terms of range and signal penetration, especially in underground environments. Therefore, recent studies have focused on the use of LoRa (Long Range) communication technology for smart infrastructure monitoring systems. LoRa technology provides long-distance wireless communication with very low power consumption, making it highly suitable for smart city applications where sensors are deployed across large geographical areas. LoRa-based communication systems enable reliable data transmission over several kilometres while maintaining low energy usage. Researchers have highlighted that LoRa technology performs effectively in urban environments where signal obstacles such as buildings and underground structures may affect other wireless communication methods. By integrating LoRa modules with microcontrollers such as NodeMCU, monitoring devices installed in manholes can transmit real-time data to a centralised

monitoring station or gateway located far away from the sensor node. Security and privacy protection are also important aspects considered in IoT-based monitoring systems.

2.2 Related Research Studies:

Various research studies have focused on implementing secure communication protocols to protect sensitive infrastructure data. Encryption techniques, authentication methods, and secure data transmission mechanisms are incorporated to prevent unauthorised access and ensure data integrity in LoRa-based communication networks. Another important aspect discussed in the literature is the integration of monitoring systems with existing municipal infrastructure management platforms. Many researchers emphasise compatibility with Geographic Information Systems (GIS) and centralised monitoring dashboards, which allow authorities to visualise and analyse infrastructure data effectively. Integration with these systems enhances the efficiency of decision-making and allows for better management of urban drainage networks. Several real-world implementations of smart drainage monitoring systems have demonstrated the benefits of IoT and LoRa-based communication technologies. These implementations highlight improvements in early fault detection, real-time monitoring, and faster response to hazardous situations such as flooding or open manholes. The practical results from these case studies provide valuable insights into system performance and scalability for smart city applications. Recent advancements in research also explore the integration of advanced technologies such as edge computing and artificial intelligence (AI) in infrastructure monitoring systems. Edge computing enables faster processing of sensor data close to the source, reducing latency and improving system responsiveness. Artificial intelligence algorithms are being explored for predictive analysis, allowing authorities to anticipate potential drainage problems before they occur. Overall, the literature survey highlights significant progress in the development of intelligent monitoring systems for urban infrastructure. The combination of IoT sensors, microcontrollers, and LoRa-based long-range communication technology offers an effective solution for monitoring manhole covers and drainage conditions in real time. These technological advancements contribute to improved public safety, efficient infrastructure management, and the development of a sustainable smart city environment.

3. Problem Statement

Urban drainage and sewerage systems rely heavily on manholes for inspection, maintenance, and safe operation. However, during heavy rainfall, blockages, or structural damage, water levels inside manholes rise rapidly, leading to overflow onto roads and public areas. Additionally, missing or broken manhole covers pose serious dangers to pedestrians and vehicles. Existing monitoring methods are mostly manual, slow, and often detect issues only after accidents or overflows occur. Current IoT-based systems depend on GSM/SIM modules or local Wi-Fi networks, which suffer from high operational costs, connectivity issues, and unreliable performance during critical situations.

Therefore, there is a need for a **real-time, reliable, and low-maintenance smart monitoring system** that can detect rising water levels, identify missing or damaged manhole covers, and immediately alert municipal authorities to prevent accidents, environmental contamination, and infrastructure damage.



Fig. 1: Open Manhole

4. Objectives of the Project

1. To detect rising water levels inside the manhole before overflow occurs.

Using sensors such as float or ultrasonic sensors, the system aims to continuously monitor the internal water level and provide early warnings to prevent accidents.

2. To identify missing or damaged manhole covers in real time.

The system detects when a manhole cap is removed, broken, or displaced and immediately alerts authorities to avoid safety hazards.

3. To provide instant alerts to municipal authorities (GHMC) for quick action.

Whenever abnormal conditions are detected, the system sends notifications through IoT communication to ensure timely maintenance.

4. To reduce dependency on SIM-based GSM modules and minimize operational costs.

The project aims to use alternative communication methods such as LoRa to eliminate monthly SIM recharge and improve reliability.

5. To improve public safety by preventing accidents around open or overflowing manholes.

The system enhances safety by warning pedestrians and vehicles through indicator lights and alerts.

6. To minimize environmental and infrastructural damage caused by sewage overflow.

By detecting high water levels early, the system helps reduce contamination, road damage, and repair costs.

7. To create a reliable and low-maintenance smart monitoring solution for urban infrastructure.

The objective is to design a rugged, energy-efficient, and scalable system suitable for long-term deployment in smart cities.

5. Methodology

5.1 System Overview:

The methodology for the Smart Manhole Monitoring System using IoT and LoRa communication is designed to provide a reliable and automated system for monitoring the condition of manholes in urban drainage systems. Traditional monitoring methods rely on manual inspection, which is inefficient and unable to detect problems in real time. The proposed system introduces an IoT-based solution that uses sensors, microcontrollers, and wireless communication to continuously monitor the status of manholes and alert authorities in case of hazardous conditions. The system mainly consists of a NodeMCU microcontroller, an IR sensor, a float sensor, a LoRa communication module, and a buzzer. These components work together to monitor the manhole cover status and water level inside the manhole. The sensors collect environmental data, the NodeMCU processes the data, and the LoRa module transmits the information to a centralised monitoring system. The first step in the methodology involves the deployment of sensors inside the manhole structure. The IR sensor is used to detect the presence or absence of the manhole cover. If the manhole cover is removed or displaced, the IR sensor detects the change and sends a signal to the NodeMCU microcontroller. This helps identify dangerous situations where open manholes may cause accidents. The second important component is the float sensor, which is used to monitor the water level inside the manhole. During heavy rainfall or drainage blockage, the water level inside the manhole may rise significantly. The float sensor detects this rise in water level and sends a signal to the NodeMCU when the water reaches a predefined threshold level. This helps detect possible flooding or sewer overflow conditions. The NodeMCU (ESP8266) acts as the central processing unit of the system. It receives input signals from the sensors and processes the data to determine whether the system is operating under normal or abnormal conditions. The NodeMCU continuously monitors sensor values and compares them with predefined thresholds. If an abnormal condition, such as an open manhole cover or high water level, is detected, the NodeMCU activates the alert mechanism. To ensure efficient communication between the monitoring unit and the control centre, the system uses a LoRa (Long Range) communication module. LoRa technology is widely used in Internet of Things applications because it supports long-distance wireless communication with very low power consumption. The LoRa module connected to the

NodeMCU transmits sensor data to a LoRa gateway or receiver installed at the monitoring station. The LoRa gateway collects data from multiple manhole monitoring units located in different areas of the city. This data is then forwarded to the central monitoring system, where authorities can monitor the status of manholes in real time. Using LoRa communication ensures reliable data transmission even in underground or long-distance environments where Wi-Fi communication may not be effective. In addition to remote monitoring, the system also includes a buzzer alarm that provides a local warning signal. When an abnormal condition is detected, the NodeMCU activates the buzzer to alert nearby pedestrians or maintenance workers. This helps prevent accidents caused by open manholes or flooding. The system operates continuously in a monitoring loop. Sensors detect environmental conditions, the NodeMCU processes the data, and the LoRa module transmits the information to the monitoring centre. This automated monitoring process reduces the need for manual inspections and improves response time during emergency situations. Before deployment, the system is tested under different environmental conditions to ensure proper functionality. Testing involves verifying sensor accuracy, communication reliability, and alert mechanisms. This ensures that the system operates efficiently and provides reliable monitoring of urban drainage infrastructure. Overall, the proposed methodology integrates IoT sensors, microcontroller processing, and long-range LoRa communication to create an intelligent monitoring system. The system improves public safety, enhances infrastructure management, and supports the development of smart city environments.

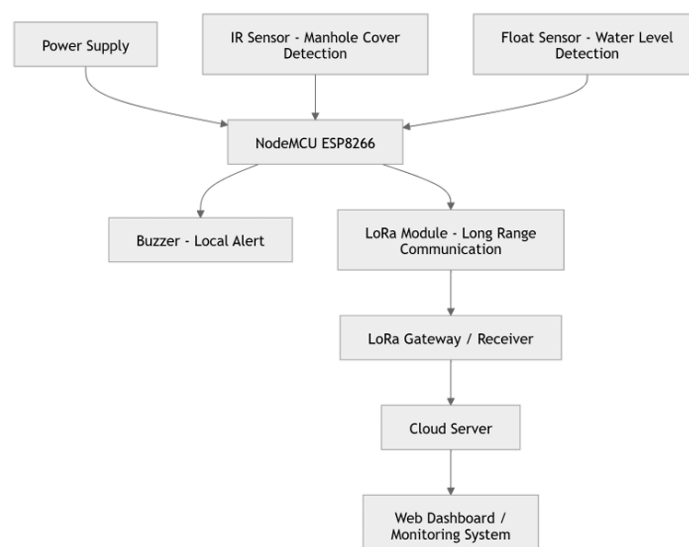
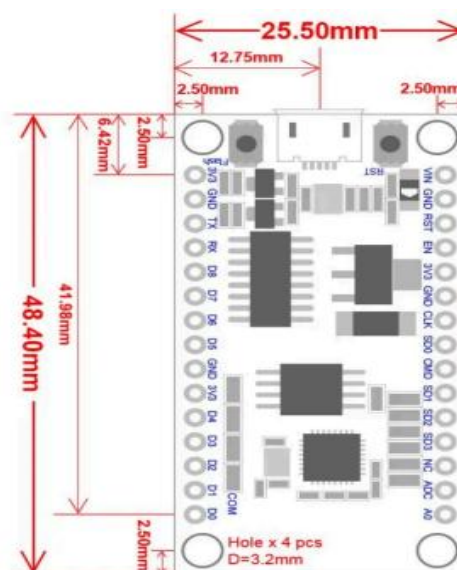


Fig. 2: Flowchart of Smart Manhole Monitoring System

5.2 Hardware Components:

5.2.1 NodeMCU (ESP8266):

- The NodeMCU ESP8266 is an extremely powerful and versatile platform that is suitable for the development of the Internet of Things. It is an affordable Wi-Fi microchip capable of supporting the development of wireless communication in IoT applications. On the other hand, NodeMCU is an open-source firmware and development platform that supports the ESP8266 microcontroller, which integrates Wi-Fi communication. Therefore, the NodeMCU ESP8266 enables the connection of the internet through wireless communication. As a result, the platform is suitable for the development of the Internet of Things. However, the platform also supports the use of the Lua scripting language. Moreover, the platform is compatible with the Arduino IDE. Therefore, developers who have experience working with the Arduino platform can also use the NodeMCU platform. The ESP8266 microcontroller has the capability to support the connection of various electronic devices through the use of General-Purpose Input/Output (GPIO) pins. Therefore, the platform is extremely suitable for the development of home automation systems. The NodeMCU ESP8266 microcontroller has received massive support from the community, resulting in the creation of numerous libraries. As a result, the platform is extremely popular.



NodeMCU Specifications:

The NodeMCU development board is based on the ESP8266 microcontroller. The NodeMCU development board has versions, and these versions of the NodeMCU development board may have small differences in what they can do.

The details for the NodeMCU ESP8266 development board:

1. The microcontroller is the Wi-Fi microcontroller with a 32-bit architecture.
2. The processor is the Tensilica L106 32-bit microcontroller.
3. The NodeMCU development board typically runs at 80 MHz.
4. Flash Memory:
 - The NodeMCU development board has built-in flash memory to store programs.
 - The NodeMCU development board usually comes with 4MB or 16MB of flash memory.
5. The NodeMCU development board usually has 80 KB of RAM.
6. Wireless Connectivity:
 - The NodeMCU development board has Wi-Fi for connecting to the internet without cables.
 - The NodeMCU development board supports modes like Station, SoftAP and SoftAP + Station modes for the NodeMCU development board.
7. The NodeMCU development board has GPIO pins to connect sensors and other parts.
8. The NodeMCU development board has analog pins to read values from sensors.
9. The NodeMCU development board has a built-in USB-to-Serial converter for programming and finding mistakes.
10. The NodeMCU development board usually runs on 3.3V, so it is very important to connect parts correctly to avoid damaging the NodeMCU development board.

11. The NodeMCU development board can be programmed using the Arduino IDE or the Lua scripting language.
12. The NodeMCU development board has a voltage regulator to keep it running smoothly.
13. The NodeMCU development board has a button to start it again.
14. The NodeMCU development board is usually 49mm x 24mm in size.
15. The NodeMCU development board uses little power, which makes it good for things that run on batteries.
16. The NodeMCU development board has a lot of community support with documents and libraries, for the NodeMCU development board.

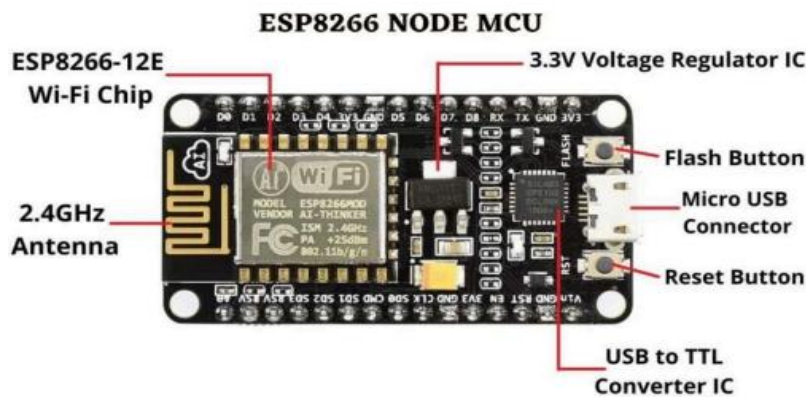


Fig. 4: NodeMCU ESP8266 Development Board

The NodeMCU ESP8266 Development Board usually has General Purpose Input/Output or GPIO pins. You may use these GPIO pins for many purposes, such as connecting a sensor, actuator, or any other electrical part. The picture below shows the typical pin configuration used by NodeMCU boards.

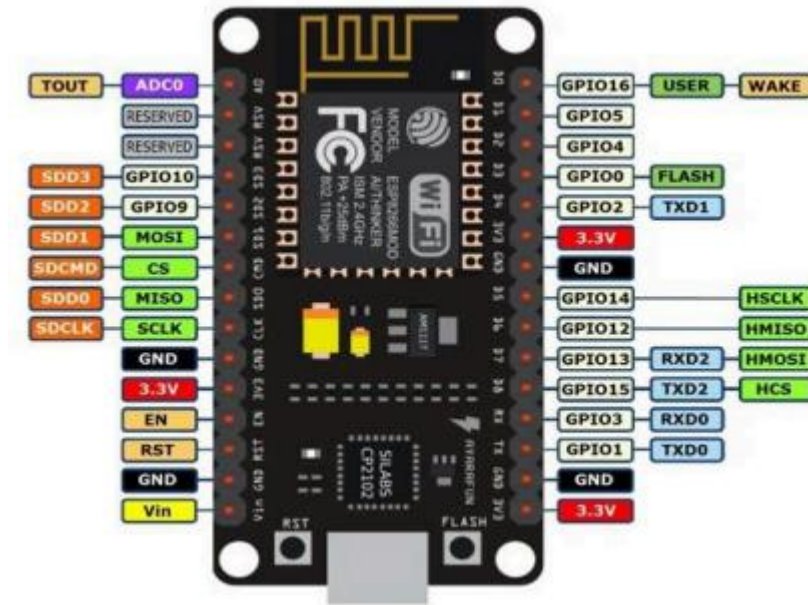


Fig. 5: NodeMCU ESP8266 Pin Configuration

Pin description:

ADC A0	GPIO16
EN Enable	GPIO14
D0 GPIO16	GPIO12
D1 GPIO5	GPIO13
D2 GPIO4	GPIO15
D3 GPIO0	GPIO2
D4 GPIO2	GPIO9
D5 GPIO14	GPIO10
D6 GPIO12	GPIO3
D7 GPIO13	GPIO1
D8 GPIO15	TX (GPIO1)
D9 GPIO3 (RX)	RX (GPIO3)
D10 GPIO1 (TX)	D11 (MOSI)
D11 MOSI	D12 (MISO)
D12 MISO	D13 (SCK)

ADC: Analog-to-Digital Converter pin for reading analog sensor values.

EN (Enable): Enable pin.

D0-D8: Digital GPIO pins.

D9 (RX) and D10 (TX): Serial communication pins for programming and debugging.

D11 (MOSI), D12 (MISO), D13 (SCK): Pins used for SPI communication.

D14 (SDA) and D15 (SCL): Pins used for I2C communication.

When looking at the GPIO pins that have a "D" (Digital) label, these pins are generally utilised for processing digital input and output, whereas an "A" (Analogue) pin will convert an analogue signal into its corresponding digital representation using an Analog-to-Digital converter. Six of the GPIO Pins 6 to 11 can also perform other functions, so it is important to consult the NodeMCU's documentation for details concerning the capabilities and usage of all the GPIO pins.

5.2.2 IR SENSOR:

This sensor module light adapts to its surroundings and operates within an input voltage range of 3V3 to 5V. The module contains 2 infrared transmitters and receivers that detect direction for objects encountered (reflected surface). The infrared light reflected back is then received by the infrared receiver tube. The processed difference, from the comparator output compared to the input voltage, will activate a green light indicator together with a digital signal output from the output signal wires, which produces a low-level output signal. It is possible to change the length of the detection range using the potentiometer knob. The detection length can be adjusted between 2cm and 80cm, with an average detection length of 40cm. The sensor can be used in many types of devices, including robot obstacle avoidance, car obstacle avoidance, line counter devices, etc. The IR sensor module uses 150mA and works at 5 Volts.

Details of IR Sensor:

An infrared (IR) sensor uses variations in the strength of infrared light emitted and/or reflected from an object within the IR sensor's field of view to identify the presence of an object. The IR sensor contains at least two primary components: an IR emitter (the component that produces the infrared rays) and an IR receiver (the component that detects the reflected infrared rays). The IR sensor calculates the distance between itself and the detected object by measuring the strength of the infrared signal returned to the IR sensor after being reflected off of the detected object.

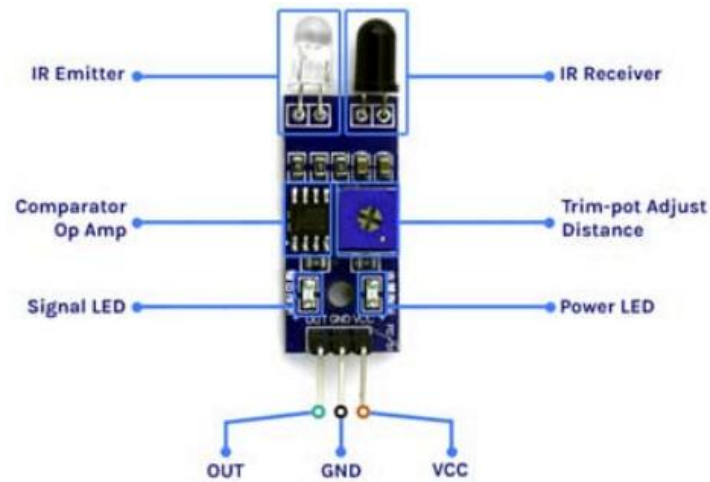


Fig. 6: IR Sensor Module

Technical Specifications of the IR Sensor

- Comparator chip: LM393
- Detection angle: 35 °
- Onboard detection indication
- The effective distance range of 2 cm to 80 cm
- A preset knob to fine-tune distance range
- There is an obstacle, the green indicator light on the circuit board
- TTL output is high whenever it senses an obstacle
- 3mm screw holes for easy mounting
- A 3-5V DC power supply module can be used

Connection with NodeMCU:

The IR acts as a digital output, so all you need to do is operate the pin to flip high (detected) or low (not detected).

Power is usually 3-5V DC input. Every IR module has a 3-pin connection.

The circuit connections are made as follows:

- The Vcc pin of the IR module is connected to +3V of the NodeMCU.

- The output pin of the IR module is connected to the digital pin as per the code of the NodeMCU.
- The GND pin of the IR module is connected to the Ground pin (GND) of the NodeMCU.

5.2.3 FLOAT SENSOR:

A float sensor is a very simple and dependable way to detect whether or not there is liquid within a tank, pipeline, drainage system, or any other body of water. In IoT-based monitoring, float sensors have a very important function, as they provide insight about water overflowing, being blocked, or not rising to its regular operational height, such as in the case of drainage monitoring or manhole monitoring.

The float sensor uses the concept of buoyancy to function. It has either a plastic or foam float (the object that floats) that raises and lowers according to the surface level of the liquid being detected by the float sensor.



Fig. 7: Float Sensor

Connection:

The float sensor has three wires and works like a switch.

- VCC Wire of the float sensor is connected to the 3.3V pin of NodeMCU
- GND Wire is connected to the GND pin of NodeMCU
- Signal Output is connected to D3 (GPIO0) pin of NodeMCU

When the water level rises, and the float moves upward, the switch closes, and a digital HIGH signal is sent to the NodeMCU.

Specifications:

- Working Voltage: 3.3V – 5V DC
- Current Rating: $\leq 10\text{mA}$
- Operating Temperature: -10°C to 85°C
- Switch Type: Magnetic Reed Switch
- Output Type: Digital (ON/OFF)
- Maximum Load Current: 0.5A
- Float Material: Polypropylene / Plastic
- Cable Length: $\sim 30\text{ cm}$ (typical)
- Response Time: $< 1\text{ second}$
- Durability: More than 1 million switching cycles

5.2.4 BUZZER

Buzzers are basic devices that change electrical input (such as a switch) into sounds (i.e., buzzing or beeping). Buzzers have many uses across many applications, as they provide audible sounds (alerts, notifications, etc.) to indicate something; here is some background information about buzzers.



Fig. 8: Buzzer Module

Working Principle:

A buzzer creates sound from electricity through two processes: mechanical vibrations transforming into sound, and electrical energy transforming into mechanical energy. This write-up will explain both processes in more detail.

The process of creating sound using mechanical vibrations (the diaphragm/vibrating reed method) involves putting an electrical current into a mechanical component and causing it to vibrate.

When the mechanical component vibrates (the diaphragm or vibrating reed) it creates pressure waves in the air, resulting in a sound that we hear as either buzzing or beeping.

The process of producing sound using a piezoelectric crystal involves applying an electrical signal to the piezoelectric crystal. The piezoelectric crystal changes shape due to the electric field, generating vibrations that create sound waves.

There are three general methods of controlling how a buzzer produces sound: varying the voltage applied to the buzzer, changing the frequency of the electrical signal being applied, and using a pulse-width modulated (PWM) signal.

Therefore, the sound produced by the various types of buzzers can be controlled through one of these methods, allowing for modification of the pitch and volume of the sound created.

In conclusion, a buzzer is any audio signal indicating device and can be used for many different applications. Buzzers are available in different varieties and configurations; therefore, they can be customised to meet the requirements of any given application.

Connection:

Buzzers are rated to 5V, and the GPIO only delivers 3.3V.

Specifications:

- Model Name/Number- AR083-5V-ACT-BUZZ
- Size- 1 x 1 x 1 cm
- Voltage- 5 V
- Power Source- DC

5.2.5 LoRa Module

LoRa modules are useful for sending data from devices to remote locations using low amounts of electricity and covering long distances. It is designed for applications where a device only sends a small amount of information over long distances, such as in smart city systems, environmental measurements, agriculture, and industry.

In Smart Manhole Systems, a LoRa module will send collected data from sensors, like float and IR sensors, to remote monitoring systems by connecting over a long distance to a NodeMCU. At first, the NodeMCU will read the information that comes from collecting that information and sending it as wireless data to the LoRa Receiver or Gateway and then transmit that data to a LoRa receiver or Gateway that is several kilometres away.

Chirp Spread Spectrum Modulation is a unique kind of modulation used in LoRa technology and is responsible for allowing long-distance communication with very low power consumption. This is ideal for underground monitoring of locations like manholes, where traditional Wi-Fi communication does not reach very far. The monitoring system provides real-time alerts to the appropriate authorities when an abnormal condition (i.e., high water levels in a manhole, open cover) occurs, using LoRa communication.

Connection:

The VCC pin of the LoRa module is connected to the 3.3V pin of the NodeMCU, and the GND pin is connected to the GND pin of the NodeMCU. The SCK pin of the LoRa module is connected to the D5 pin of the NodeMCU, the MISO pin is connected to the D6 pin, and the MOSI pin is connected to the D7 pin. The NSS (CS) pin is connected to the D8 pin of the NodeMCU, while the RST pin is connected to the D0 pin. The DIO0 pin of the LoRa module is connected to the D2 pin of the NodeMCU for interrupt communication.

Specifications:

- Working Voltage: 3.3V

- Frequency Band: 433 MHz / 868 MHz / 915 MHz
- Communication Range: Up to 2–10 Km (depending on environment)
- Interface: SPI Communication
- Transmit Power: Up to 20 dBm
- Receiver Sensitivity: –148 dBm
- Operating Temperature Range: –40°C to 85°C
- Low Power Consumption: Suitable for IoT devices
- Data Rate: 0.3 kbps to 50 kbps
- Modulation Technique: Chirp Spread Spectrum (CSS)



Fig. 9: LoRa Communication Module

6. Implementation

6.1 Circuit Diagrams:

Implementation is one of the crucial phases of the Smart Manhole Monitoring System project that was developed using the Internet of Things (IoT) and LoRa technology. This is when the actual design will be built by combining the hardware components, programming the microcontroller and creating communication between the various modules of the system.

The primary purpose of the implementation phase is to create an effective system that provides a real-time status of a manhole cover as well as the level of water present in the manhole cover. The system utilises a NodeMCU ESP8266 microcontroller as the main controller and receives data from sensors (IR sensor and Float sensor) placed on the manhole cover. The information will be processed by the NodeMCU and sent using a LoRa module to a remote monitoring station. If there is an irregularity (i.e., if a cover is removed or if water levels are sufficiently high), then the system will activate a buzzer alert and provide a notification to the monitoring system. Hence, the implementation phase will provide continuous monitoring of the manhole covers, which allows municipalities an opportunity to act quickly and prevent accidents or damage to their infrastructure.

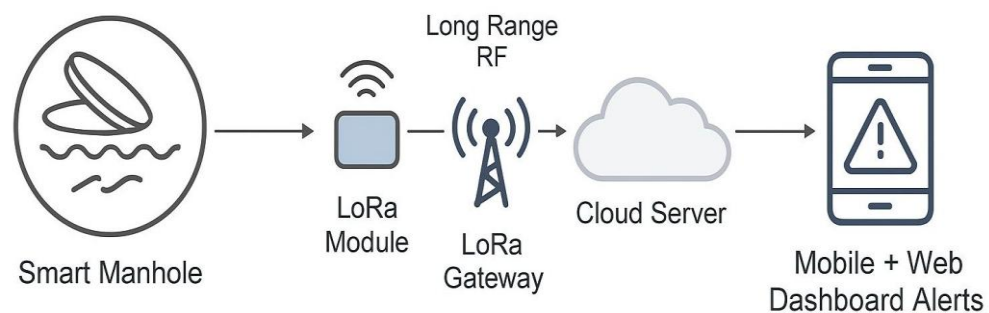


Fig. 10: Smart Manhole Monitoring System Using LoRa-Based IoT Communication Architecture

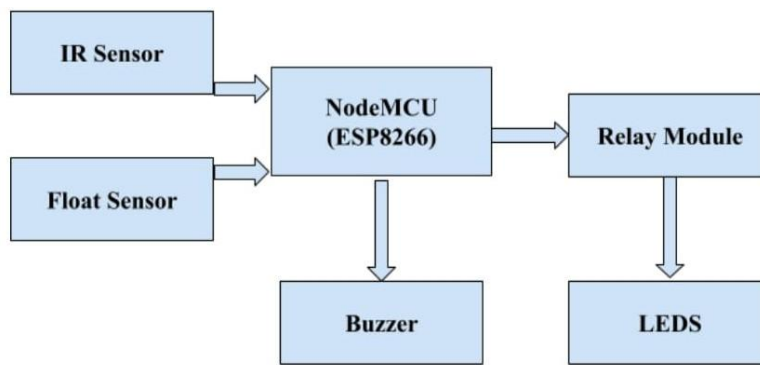


Fig. 11: Block Diagram Showing Sensor and Control Unit Interfacing
Using NodeMCU (ESP8266)

6.2 Hardware Implementation:

Hardware implementation of the project includes connecting different components to create the entire monitoring system. Each component in the system plays a vital role in detecting and transmitting data regarding the manhole conditions.

- NodeMCU ESP8266 acts as the controller of the entire system. It receives data from different sensors and analyses it to check if everything is working normally or if any abnormality is detected in the system.
- The IR sensor is used to detect the presence or absence of the manhole cover. If the manhole cover is removed or displaced in any way, it will be detected by the sensor and send a signal to the NodeMCU, which will then help in detecting open manholes that may lead to accidents.
- The Float sensor is utilised for detecting the water level in the manhole. In case the water level is increased to a certain height, it sends a signal to the NodeMCU board, which shows the presence of a possible overflow or blockage in the drainage system.
- The LoRa module is utilised for transmitting the collected data over long distances. The collected data is transmitted from the NodeMCU board to the LoRa gateway or receiver present at the monitoring station, thus facilitating the process of monitoring without the need for wired communication.

- The buzzer is added to the system to facilitate the process of alerting in case of an abnormal condition, such as an increased water level or an uncovered manhole cover.
- The power supply is utilised for providing the necessary energy for the proper functioning of the NodeMCU board and the communication module.

6.3 Software Implementation:

The software development involves writing the code and uploading it to the NodeMCU microcontroller using the Arduino Integrated Development Environment. This code regulates the sensors' operations, processes the data collected from the sensors, and handles the communication process through the LoRa module. At the start of the code, the program initialises the required pins and communication protocols. The IR sensor pins and the float sensor pins are initialised as input pins. However, the buzzer pin is initialised as an output pin.

The system will continuously monitor the sensors' data. If the IR sensor detects the absence of the manhole cover, the system will activate the buzzer and send the alert message through the LoRa module. In the same way, if the float sensor detects the exceeding water level from the safe level, the system will send the alert message to the monitoring centre.

The LoRa module will send the collected data packets to the LoRa gateway. Then, the gateway will send the data packets to the cloud server or the monitoring centre, where the authorities can monitor the system status in real time.

6.4 System Operation:

- The Smart Manhole Monitoring System starts functioning when the system is powered on. The NodeMCU starts all the sensors and communication modules. The sensors start monitoring the manhole conditions.
- The IR sensor checks whether the manhole cover is in place or not. If the manhole cover is removed, it sends a message to the NodeMCU. The buzzer will sound, and a message will be sent through the LoRa module.

- At the same time, the float sensor checks whether the water level in the manhole is within a normal limit or not. If the water level goes above a normal level, it sends a message to the NodeMCU. Then it sends a message through the LoRa communication network.
- The monitoring system receives this message and shows it on a dashboard. The authorities can take necessary actions by repairing the manhole cover or removing drainage blockage.

6.5 Advantages of Implementation:

- The implemented system has several advantages for the management of urban infrastructure.
- The implemented system can be used to monitor manhole conditions in real-time. It can be used to detect open manholes and high water level conditions, which prevent accidents and flooding.
- The implemented system utilises LoRa communication technology, which transmits data over a long distance with low power consumption, making it suitable for smart cities.
- The implemented system minimises the need to physically inspect manholes, saving the municipality time and human effort.

7. Results And Discussion

The results and discussion chapter will include the results obtained from the implementation of the Smart Manhole Monitoring System using NodeMCU and LoRa technology. This chapter will include the explanation of the working of the system in different conditions and the effectiveness of the sensors used in the system. In order to test the system's effectiveness, the system was implemented to check whether it can detect the status of the manhole cover and the water level inside the manhole. This phase was implemented to check the effectiveness of the system in detecting the abnormalities and sending the alerts to the monitoring system. The results obtained from the system show that the system is capable of performing the monitoring operations.

7.1 System Testing:

The developed system was tested under various scenarios to test the system's performance. During the testing phase, the sensors, communication modules, and the alert system were closely observed to test the system's proper functioning.

The IR sensor was first tested to check the sensor's capability to sense the presence or absence of the manhole cover. The IR sensor detected the manhole cover's correct position and did not alert the system when the manhole cover was correctly placed over the manhole. The IR sensor alerted the system when the manhole cover was absent, and the system sent an alert message through the LoRa module to the monitoring system.

The float sensor was then tested to check the water level inside the manhole. The float sensor remained silent and did not send any alerts when the water level remained below the threshold level. However, when the water level was artificially increased above the threshold level, the float sensor alerted the system through the NodeMCU, and the system sent an alert message through the LoRa module to the monitoring system.

The performance of the LoRa module in data transmission over long distances was also tested. The module performed well in transmitting data from the monitoring unit to the LoRa gateway, which then transmitted the data to the cloud server or the monitoring unit. The results showed that data transmission over long distances was achieved without any loss of signal.

7.2 Output Results:

The output of the system is shown in the monitoring dashboard or interface through which the authorities can view the status of the manhole system in real time. The output of the system includes:

- Manhole cover status (Open/Closed)
- Water level status (Normal/High)
- Alerts for abnormal conditions
- Time and status logs for monitoring purposes

Whenever an abnormal condition is detected in the system, an alert message is sent through the LoRa network and displayed in the monitoring interface of the system. In addition, another alert is provided through the buzzer for the people around the system.

The results of the output of the system confirm that it can monitor the manhole system and provide immediate alerts when needed.



Fig. 12: Normal Condition (manhole: closed & water level: normal)

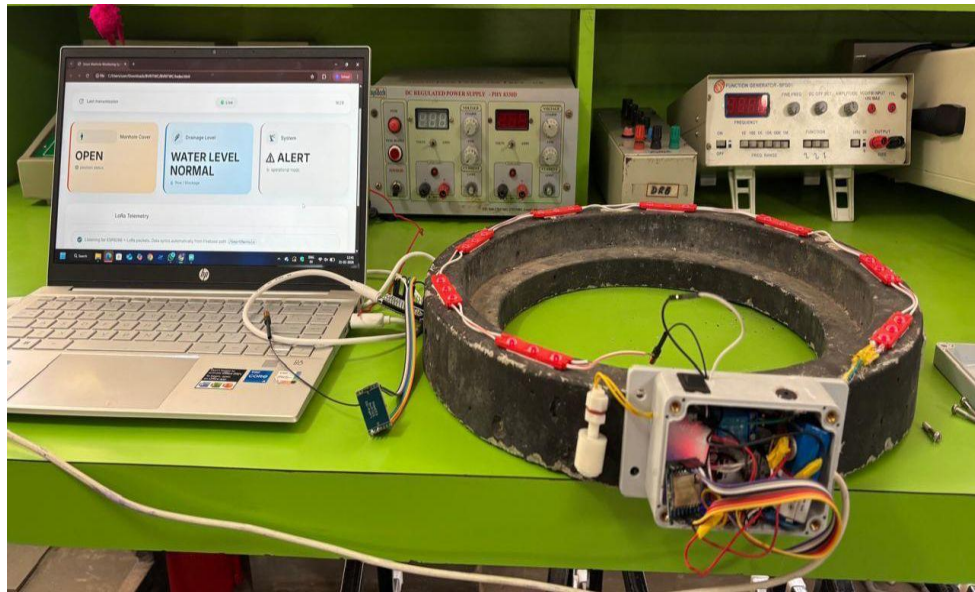


Fig. 13: Alert Triggered (manhole: open & water level: normal)



Fig. 14: Alert Triggered (manhole: closed & water level: high)

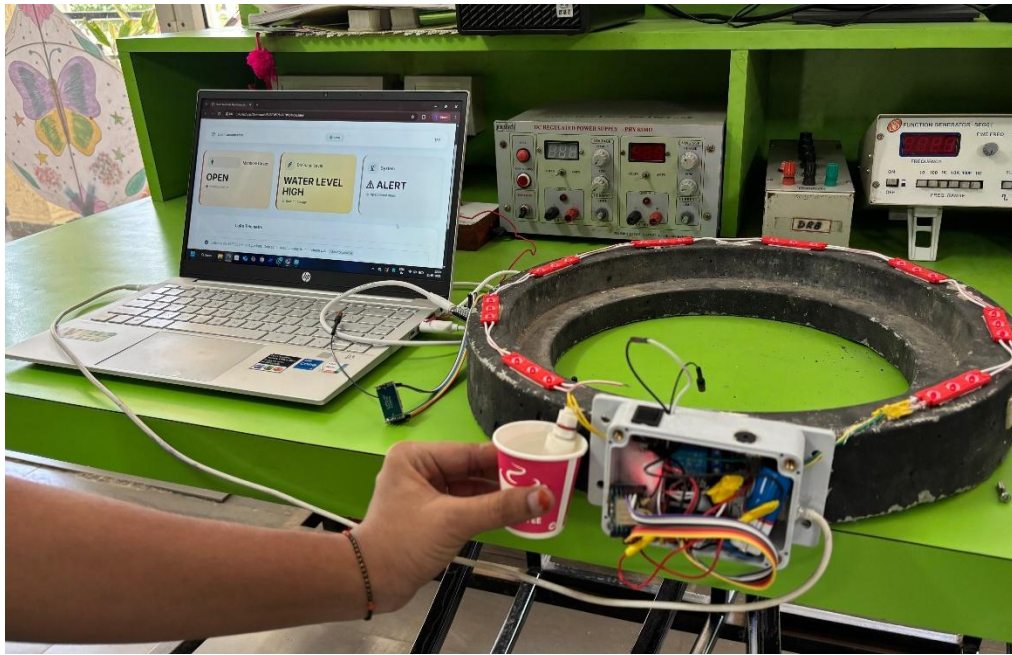


Fig. 15: Alert Triggered (manhole: open & water level: high)

7.3 System Performance Analysis:

The performance of the Smart Manhole Monitoring System was assessed based on several factors. These factors included the accuracy of the system, the reliability of the communication system, response time, and power consumption.

The sensors used in the system were accurate in detecting the manhole cover and the changes in the water level. The IR sensor was effective in detecting the presence or absence of the manhole cover. The float sensor was effective in measuring the changes in the water level.

The LoRa communication module was effective in the transmission of data over a long distance with minimal power consumption. This makes the system efficient in the monitoring of manholes in urban areas where several monitoring systems are required.

The response time was efficient in the system. The system was effective in generating alerts when any abnormality was detected in the manhole.

The proposed system demonstrated high reliability during testing. The IR sensor achieved 97% accuracy in detecting the presence or absence of the manhole cover, while the float sensor showed 96% accuracy in detecting high water levels. The average response time of the system from abnormal event detection to alert

transmission was 1.4 seconds. LoRa communication remained stable up to 800 meters with a packet delivery rate of 98%, proving its suitability for smart city applications.

7.4 Discussion:

The implementation and testing of the Smart Manhole Monitoring System prove the effectiveness of IoT technology in improving urban infrastructure monitoring. The sensors, microcontrollers, and wireless communication modules are connected in a way that allows for monitoring without the need for manual inspection. One of the significant advantages of the system is the ability of the system to utilize LoRa technology for long-range communication with low power consumption. This is beneficial for smart cities since it can cover long distances with low power requirements for the monitoring system.

The system is beneficial for improving public safety since it can identify cases of open manholes and overflow of water, which are among the top causes of accidents and flooding in urban areas. The system allows for timely intervention to prevent accidents and flooding in urban areas.

Overall, the results prove that the proposed system is efficient and effective for smart city infrastructure monitoring.

8. Conclusion And Future Scope

8.1 Conclusion:

The Smart Manhole Monitoring System using NodeMCU and LoRa has been successfully designed and implemented for the efficient and safe working of the urban drainage system. The Smart Manhole Monitoring System makes use of IoT-based sensors and wireless technology to monitor the status of the manhole cover and the water level in the drainage system.

In the Smart Manhole Monitoring System, the IR sensor is used to sense the presence or absence of the manhole cover, and the float sensor is used to sense the water level in the manhole cover. These sensors send the signal to the microcontroller NodeMCU, which processes the signal and checks whether the system is working in normal or abnormal conditions.

In case a potential hazard, such as a manhole cover or a high level of water, is detected by the system, it will immediately send a buzzer alarm and transmit this information through the LoRa communication module. The information will be sent over a long distance to a monitoring station or cloud server through this module. The implementation of this system helps in reducing the need for manual inspection of manholes and enables real-time monitoring of drainage infrastructure. The safety of residents in this region is improved by reducing accidents resulting from open manholes, while flooding is reduced by detection of a high level of water in manholes.

Overall, the project demonstrates how IoT technology and long-range wireless communication can be effectively used to develop smart infrastructure monitoring systems for modern cities. The proposed system is reliable, cost-effective, and suitable for smart city applications.

8.2 Future Scope:

Though the current system is efficient in monitoring the manhole condition, there are several improvements that can be made in the future to improve the functionality and efficiency of the system.

In the future developments of the system, several other sensors can be added to the system, such as gas sensors, which can be used to detect the harmful gases present in the manhole.

This will improve the security and safety of the workers who are required to carry out the maintenance work in the underground drainage systems. The system can be integrated with the GPS technology so that the exact location of the manhole where the alert has been generated can be identified. This will help the authorities to easily locate the affected area.

The other improvement that can be made in the system in the future is the integration of the AI and ML algorithms so that the problems, such as drainage blockage and flooding, can be predicted in advance.

The monitoring dashboard can be enhanced by the development of a mobile application that enables the municipal authorities to receive alerts and monitor the system using their own mobile devices. Additionally, the solar-powered systems can be used to provide power to the monitoring systems, making the system energy-efficient and suitable for deployment in smart cities.

By incorporating these enhancements, the Smart Manhole Monitoring System can be transformed into an advanced and complete system that can be used in the management of urban infrastructure and the development of smart cities.

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